

Problem

The step response of an RLC circuit is given by

$$\frac{d^2i}{dt^2} + 2\frac{di}{dt} + 5i = 30$$

Given that $i(0) = 18$ A and $di(0)/dt = 36$ A/s, solve for $i(t)$.

Step-by-step solution

Step 1 of 3

The step response of the RLC circuit is,

$$\frac{d^2i}{dt^2} + 2\frac{di}{dt} + 5i = 30 \quad \dots\dots (1)$$

The initial current conditions are, $i(0) = 18$ A and $\frac{di(0)}{dt} = 36$ A/s.

Apply Laplace Transform to the equation 1.

$$\left(s^2 I(s) - si(0) - \frac{di(0)}{dt} \right) + 2(sI(s) - i(0)) + 5I(s) = \frac{30}{s}$$

Substitute 18 A for $i(0)$ and 36 A/s for $\frac{di(0)}{dt}$.

$$s^2 I(s) - 18s - 36 + 2(sI(s) - 18) + 5I(s) = \frac{30}{s}$$

$$(s^2 + 2s + 5)I(s) - 18s - 72 = \frac{30}{s}$$

$$s(s^2 + 2s + 5)I(s) - 18s^2 - 72s = 30$$

$$I(s) = \frac{18s^2 + 72s + 30}{s(s^2 + 2s + 5)}$$

Step 2 of 3

Take partial fractions.

$$I(s) = \frac{A}{s} + \frac{Bs + C}{(s^2 + 2s + 5)} \dots\dots (2)$$

$$18s^2 + 72s + 30 = A(s^2 + 2s + 5) + (Bs + C)s$$

Comparing the coefficients on both sides to calculate A , B , and C .

$$18 = A + B \dots\dots (3)$$

$$72 = 2A + C \dots\dots (4)$$

$$30 = 5A$$

$$A = \frac{30}{5}$$

$$= 6$$

Substitute 6 for A in equation (3).

$$18 = 6 + B$$

$$B = 12$$

Substitute 6 for A in equation (4).

$$72 = 2(6) + C$$

$$C = 60$$

Step 3 of 3

Substitute 6 for A , 12 for B , and 60 for C in equation (2).

$$I(s) = \frac{6}{s} + \frac{12s + 60}{(s^2 + 2s + 5)}$$

$$= \frac{6}{s} + \frac{12s}{(s^2 + 2s + 5)} + \frac{60}{(s^2 + 2s + 5)}$$

By applying Inverse Laplace Transform to equation 4, we get

$$L^{-1}[I(s)] = L^{-1}\left[\frac{6}{s}\right] + L^{-1}\left[\frac{12s}{s^2 + 2s + 5}\right] + L^{-1}\left[\frac{60}{s^2 + 2s + 5}\right]$$

$$= 6L^{-1}\left(\frac{1}{s}\right) + 12L^{-1}\left(\frac{s}{s+1} \times \frac{s+1}{(s+1)^2 + 2^2}\right) + 30L^{-1}\left(\frac{2}{(s+1)^2 + 2^2}\right)$$

$$= 6L^{-1}\left(\frac{1}{s}\right) + 12L^{-1}\left(\left(1 - \frac{1}{s+1}\right) \times \frac{s+1}{(s+1)^2 + 2^2}\right) + 30L^{-1}\left(\frac{2}{(s+1)^2 + 2^2}\right)$$

$$= 6L^{-1}\left(\frac{1}{s}\right) + 12L^{-1}\left[\frac{s+1}{(s+1)^2 + 2^2}\right] - 6L^{-1}\left[\frac{2}{(s+1)^2 + 2^2}\right] + 30L^{-1}\left(\frac{2}{(s+1)^2 + 2^2}\right)$$

Apply inverse Laplace transform.

$$i(t) = 6u(t) + 12e^{-t} \cos(2t)u(t) - 6\sin(2t)u(t) + 30\sin(2t)u(t)$$

$$= 6u(t) + 12e^{-t} \cos(2t)u(t) + 24\sin(2t)u(t)$$

$$= 6[1 + 2e^{-t} \cos(2t) + 4\sin(2t)]u(t) \text{ A}$$

Thus, the $i(t)$ is $\boxed{6[1 + 2e^{-t} \cos(2t) + 4\sin(2t)]u(t) \text{ A}}$.